MIL-P-45651B(MU)
30 APRIL 1964

SUPERSEDING
MIL-P-45651A(MU)
17 FEBRUARY 1964

MILITARY SPECIFICATION

PERISCOPES, TANK: M31 AND M31C

1. SCOPE

1.1 This specification covers two periscopes of the same basic type—Periscopes, Tank: M31 and M31C (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MlLITARY

MIL-O-13830 —Optical Components for Fire Control Instruments; General Specification Governing -the Manufacture, Assembly, and inspection of

MIL-F-13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

MIL-P-14232 —Parts, Equipment and Tools for Ordnance Materiel, packaging of

MIL-I-45208 -Inspection System Requirements

MIL-I-45607 —Inspection Equipment, SUPply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for inspection by Attributes

MIL-STD-109 Quality Assurance Terms and Definitions

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F8289172 —Periscope, Tank: M31

F10516388 —Periscope, Tank: M31C

LSQAP 8289172 —Supplementary Quality Assurance Provisions List for Periscope, Tank: M31

LSQAP 10516388 —Supplementary Quality Assurance Provisions List for Periscope, Tank: M31C

IEL 8289172 —List of Inspection Equipment Numbers" for Periscope, Tank: M31

IEL 10516388 -List of Inspection Equipment Numbers for Periscope, Tank: M31C

FSC 1240

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PACKAGING DATA SHEET

MIL-P-14232/-Packaging of Periscopes, Tank:
P8289172 M31 and M31C

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting titer.)

3. REQUIREMENTS

3.1 Materials, Materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication. The periscope designated in the contract or purchase order shall be manufactured in accordance with the applicable drawings listed in Section 2 and all other drawings pertaining thereto.

3.3 General specifications. The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specifications MIL-O-13830 and MIL-F-13926:

3.3.1 MIL-O-13830.

(a) Cleanliness.
(b) Illuminated reticle.

3.3.2 MIL-F-13926.

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Electrical and electronic assemblies.
(e) Part identification and marking.
(f) Workmanship.

9.4 Environmental.

3.4.1 Storage temperatures. The periscope shall show no evidence of physical failure and shall meet all performance requirements of this specification at a standard ambient temperature of 60° to 90° Fahrenheit (F.) after having been exposed to and thermally stabilized at ambient temperatures of –80° and +160° F.

3.4.2 Operating temperatures. The periscope shall meet the applicable requirements of 3.6.1, 3.8.5, 3.9.6, 3.9.7, and 3.9.12.3 while exposed to and thermally stabilized at ambient temperatures of 40° and +150° F.

3.4.3 Vibration. The periscope shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ± 15 seconds. Subsequent to vibration, there shall be no evidence of physical failure and the periscope shall meet all the following requirements of this specification.

3.4.4 Line of sight retention, The line of sight of the periscope shall be retained within 0.2 mil in elevation and deflection of an original setting subsequent to the vibration specified in 3.4.3.

3.5 Cleanliness. The periscope shall meet the cleanliness requirements of Specification MIL-O-13830, and there shall be no more than 2 particles of dirt visible at area C of the mirror, when viewing through the 8-power system, nor more than 4 particles apparent in the unity power system. Each particle shall not be in excess of an area equivalent to a size 10 dig. There shall be no dirt which interferes with observation or adversely affects image quality.

Note. The following references apply throughout this specification.

Head Assembly
Area C Dwg. B8289186 (8X aperture)
Datum plane X Dwg. F8289176 (locating pads)
Datum plane Y Dwg. F8289176 (horizontal stops)
Datum plane Z Dwg, F8289173 (Body mounting surface)

Body Assembly
For {eriscope M31
Reference line Y and
Datum plane X Dwg. F8289211 (Intersection of planes)

For Periscope M31C
Reference line Y and
Datum plane X Dwg. F10516378 (Intersection of planes)

3.6 Parallax. With the parallax adjustment centered at the midpoint of the ob-

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jective travel, parallax at the optical axis shall not exceed 0.2 mil when viewing a target at 500 meters to infinity.

3.6.1 Parallax adjustment. The torque required to rotate the parallax adjustment screw shall not exceed 9 pound inches when subjected to the temperatures specified in 3.4.2.,

3.7 Sealing. The periscope head and body. when tested separately, Shall show no evidence of leakage when subjected to an internal pressure of 5 pounds per square inch gage (psig) for a period of 5 minutes. The interior atmosphere of the accepted periscope shall consist of dry nitrogen. The nitrogen shall have a dew point at least as low as −25°F,

3.8 Head assembly. With datum plane X in a vertical plane within 10 seconds, datum plane Y in a horizontal plane within 10 seconds, and side Y of the input coupling arm 8298397, referenced on Drawing F8289173, positioned 67 degrees and 30 minutes with the horizontal, the head assembly shall meet the requirements of 3.8.1 to 3.9.12.3 inclusive.

3.8.1 Collimation. An incident line of sight shall be perpendicular to datum plane X.

3.8.1.1 Light power system. The reflected line of sight when measured in area C Shall be perpendicular to datum plane Z within 0.2 mil laterallv and 0.2 mil in elevation.

3. 8.1.2 Unity power system. The reflected line of sight in the unity power viewing area shall be laterally perpendicular to datum plane X within 15 minutes and shall be in a Plane depressed 20 degrees to the horizontal within 15 minutes.

3.8.2 Plumb travel. The line of sight shall not deviate from a plumb line by more than 0.25 mil when e]evated 22 degrees and depressed 18 degrees by means of the coupling, starting with the line of sight coincident with the plumb line at the horizontal position.

3.8.3 Elevation accuracy. Each angular change at the input coupling, excluding backlash. shall cause the same angular change in the line of sight within 0.2 mil, from 18 degrees depression to 22 degrees elevation.

3.8.3,1 Backlash. Backlash shall not exceed 0.05 mil.

3.8.4 Coupling travel. Rotation of the elevation input coupling shall be limited only by contact with the travel stop pin.

3.8.5 Torque. The running torque required to rotate the coupling through maximum elevation and depression shall not exceed 7 pound inches at a standard ambient temperature of 60° to 90° F. The torque shall not change from the value obtained by more than 3 pound inches when subjected to the temperatures specified in 3.4.2 (see 6.3).

3.8.6 Resolution. The resolving power on the optical axis shall be 6 seconds of arc or less when viewing through the target viewing system (area C), and 45 seconds of arc 0r less for the remaining area.

3.9 Body assembly.

3.9.1 Collination.. The line of sight through the boresight mark shall be perpendicular to datum plane X within 0.2 mil as adjusted by the boresight knobs.

3.9.2 Elecration knob travel. The elevation boresight knob shall provide a minimum travel of 5.5 mils in elevation and 8.5 mils in depression from the collimation setting of 3.9.1.

3.9.3 A zimuth knob travel. The azimuth boresight knob shall provicle a minimum travel of 7 mil in each direction from the collimation setting of 3.9.1.

3.9.4 Reticle shift. Rotation of the boresight knobs shall not cause the reticle to shift more than 0.2 mil in a direction perpendicular to the movement of the reticle cell.

3.9.5 Backlash. Backlash in each boresight mechanism shall not exceed 0.05 mil.

3.9.6 Boresight knob disengagement. The force required to disengage the boresight

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knob lock shall not exceed 5 pounds at a standard ambient temperature of 60° to 90° F. and the temperatures specified in 3.4.2. The slip scales shall be movable at the above temperatures.

3.9.7 Boresight knob torque. The running torque required to rotate each boresight knob, when unlocked, shall not exceed 5 pound inches at a standard ambient temperature of 60° to 90° F., and the torque shall not change from the value obtained by more than 2 pound inches when subjected to the temperatures specified in 3.4.2.

3.9.8 Reticle illumination. With a 24-volt current source actuating the reticle light, the reticle marking shall appear clearly defined when observed in a dark room.

3.9.9 Image tilt. With the mounting surface vertical and reference line Y horizontal, the image of a plumb line shall be vertical within one degree.

3.9.10 Parallelism of image and reticle. With the mounting surface vertical and reference line Y horizontal, the image of a horizontal line shall be parallel to the horizontal reticle line within 15 minutes of arc.

3.9.11 Resolution. The resolving power on the optical axis shall be 6 seconds of arc or less.

3.9.12 Eyepiece focus. The diopter scale shall indicate zero within 1/4 diopter when the reticle is brought into sharp focus.

3.9.12.1 Diopter movement. The eyepiece shall be capable of movement throughout a range of +4 and -4 diopters.

3.9.12.2 Backlash. Backlash in the diopter movement shall not exceed l/8 diopter.

3.9.12.3 Diopter knob torque. The running torque required to rotate the diopter adjustment shall not exceed 7 pound inches at a standard ambient temperature of 600 to 90° F. The torque shall not change from the value obtained at standard ambient temperature by more than 3-pound inches when subjected to the temperatures specified in 3.4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-I-45208 and the component inspection requirements of Specification MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be listed in Standard MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot. Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with “Submission of Product” as specified in Standard MIL-STD-105.

4.2.2 Examination and tests. Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926, and Supplementary Quality Assurance Provisions LSQAP 8289172 for Periscope: M31, and LSQAP 10516388 for Periscope: M31C, as applicable. Examinations and tests related to section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105

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and the inspection level and sampling plans specified in table I, Classification of Defects. Examinations and tests for packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

TABLE I. Classification of defects
Use Inspection Level II of Table I
with Sampling Plan Table II-A of Standard MIL-STD-105

Critical: None.

Major AQL 0.65%	Requirement	Test Procedure
Periscope		
101. Physical failure, post vibration	3.4.3	4.5.2.1
102. Line of sight retention	3.4.4	4.5.2.2
103. Cleanliness	3.5	4.5.3
104. Parallax	3.6	4.5.4
105. Torque, parallax adjustment screw	3.6.1	4.5.5
Head Assembly		
106. Collimation	3.8.1.1, 3.8.1.2	4.5.7.1
107. Plumb travel	3.8.2	4.5.7.2
108. Elevation accuracy	3.8.3	4.5.7.3
109. Backlash	3.8.3.1	4.5.7.4
110. Coupling travel	3.8.4	4.5.7.5
111. Torque	3.8.5	4.5.7.6
112. Resolution	3.8.6	4.5.7.7
113. Collimation	3.9.1	4.5.8.1
114. Elevation knob travel	3.9.2	4.5.8.2
115. Azimuth knob travel	3.9.3	4.5.8.3
116. Reticle shift	3.9.4	4.5.8.4
117. Backlash	3.9.5	4.5.8.5
118. Foresight knob disengagement	3.9.6	4.5.8.6
119. Foresight knob torque	3.9.7	4.5.8.7
120. Reticle illumination	3.9.8	4.5.8.8
121. Image tilt	3.9.9	4.5.8.9
122. Parallelism of image and reticle	3.9.10	4.5.8.10
123. Resolution	3.9.11	4.5.7.7, 4.5.8.11
124. Eyepiece focus	3.9.12	4.5.8.12
125. Diopter movement	3.9.12.1	4.5.8.13
126. Backlash	3.9.12.2	4.5.8.14
127. Diopter knob torque	3.9.12.3	4.5.8.7
Minor None.		

4.2.3 Disposition of nonconforming p~oduct. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL-STD-105.

4.3 Control tests.

Control test	Requirement	Test procedure
301. Materials	3.1	Applicable drawings and specs.
302. Fabrication	3.2	Visual per drawings
303. General Specifications	3.3	Visual, MIL-O-13830, MIL-F-13926.

4.3.1 General. One periscope shall be selected at random as a control sample from each 100 produced, or from each month's production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

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4.3.2 Environmental. Three periscopes shall be selected at random as control samples from each 50 produced, or from each month's production, whichever occurs first. The control samples shall meet the following requirements and tests:

Control test	Requirement	Test procedure
304. Storage temperature.	3.4.1	4.5.1.1
305. Operating temperatures	3.4.2	4.5.1.2
306. Sealing	3.7	4.5.6

4.3.3 Control test failure. Should anyone item of a control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Measuring and Test Equipment" requirement of Specification MIL-I-45208 and all applicable requirements specified in Specification MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Govenent furnished inspection equipment. Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the "Measuring and Test Equipment" and "Government Furnished Material" requirements of Specification MIL-I-45208 and all applicable requirements specified in Specification MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design. All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the requirements of 4.4 and the List of Inspection Equipment Numbers, IEL 8289172 for Periscope M31 and IEL 10516388 for Periscope M31C, as applicable. Alternate designs may be substituted only as provided by "Inspection Provisions" of Specification MIL-I-45208.

4.4.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.2.1 lnspection equipment design requirements.

4.4.2.2.2 Head assemldy.

4.4.2.2.3 Special test equipment to test collimation plumb travel, elevation accuracy, and alignment (unity power arm). The device shall permit mounting the head assembly with the datum plane X in a vertical plane within 10 seconds, datum plane Y in a horizontal plane within 10 seconds, and side Y of the input coupling 67 degrees 30 minutes 0 seconds $\pm$ 10 seconds with respect to datum plane Y. Corresponding mating locating surfaces of the device shall be in complete contact with locating surfaces (datum plane X, datum plane Y, and side Y) of the head assembly,

4.4.2.2.4 Body assembly.

4.4.2.2.5 Special testing equipment to test collimation setting, reticle shift, backlash,

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image tilt, and parallelism of reticle and image. The device shall permit mounting the body assembly with datum plane X vertical within 10 seconds and reference line Y horizontal within 10 seconds. Corresponding mating locating surfaces of the device shall be in complete contact with the locating surfaces (datum plane X and referenced line Y) of the body assembly. The device shall also be equipped with a 24-volt direct current power source for energizing the reticle light.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Storage temperatures. This test shall be conducted in accordance with the conditions of 3.4.1 and the applicable Test Facilities of Specification MIL-F-13926, Upon completion of the specified exposure, the periscope shall be inspected for damage and then subjected to all the tests of this specification in accordance with 3.4.1.

4.5.1.2 Operating temperatures. This test shall be Conducted in accordance with the conditions of 3.4.2 and the applicable Test Facilities of Specification MIL-F-13926. While exposed to and thermally stabilized at the indicated temperatures, the periscope shall be subjected to the tests specified in 4.5.5.4.5.7.6, 4.5.8.6, and 4.5.8.7. The controls of the periscope shall meet the torque and force requirements for the indicated operating temperatures as specified in 3.4.2.

4.5.2 Vibration. Standard vibrating equipment capable of providing the amplitude and frequeney as specified in 3.4.3 shall be utilized for this test. The periscope shall be vibrated in a vertical plane. Unless otherwise specified the test equipnent shall be in conformance with the Test Facilities requirements of Specification MIL-F-13926.

4.5.2.1. Post vibration. Subsequent to vibration the

ined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4.3 prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts before the periscope is subjected to the remaining performance tests. Minimum visual examination shall include boresight locks, elevation input coupling, body assembly, head assembly, and eyepiece assembly. A tactile examination should also be conducted to determine looseness or improper functioning of such moving parts as the input coupling, azimuth and elevation knobs, boresight knob and lock, as well as screws, nuts, and other assembled parts which may have been loosened as a result of vibration. An audible examination shall be made by holding the periscope close to the ear and shaking the periscope by hand to try to detect loose optical and mechanical parts such as lenses, cells, retainers, prisms, adapters, or separated bonded elements, Any one of the possible defects defined above shall be cause to consider the periscope as defective unless separate defects per-hundred type sampling plans are established and approved by the Government as a part of the supplier's inspection system. Periscopes which have passed the above examinations shall be subjected to the tests of 4.5.2.2 to 4.5.8.14, inclusive, to determine overall conformance with 3.4.3.

4.5.2.2 Line of sight retention. This test shall be performed prior to vibration in accordance with 4.5.7.1, and the results noted. Subsequent to vibration the test shall again be performed and the line of sight shall be within the tolerance as specifiecd in 3.4.4.

4.5.3 Cleanliness. The periscope shall be inspected for cleanliness in accordance with Specification MIL-O-13830 and shall meet. the requirements specified. The mean diameter of a size 19 dig is equal to 0.10 Millimeter.

4.5.4 Parallax. Position the periscope. in its normal operating position. in a holding device. Adjust the boresight knobs of the body assembly to their mid-point of travel, as indicated in 3.9.2 and 3.9.3, and the parallax adjustment screw as specified. Direct a line of sight through the periscope boresight

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point to a sharply defined target located at the specified distance from the periscope window. Parallax will be recognized as any apparent movement of a reticle in relation to' the target image when the observer's head is moved from side to side or up and down. Parallax at the boresight point with respect to the target image shall not exceed the tolerance specified in 3.6. Note: 0.1 mil-Thickness of a reticle line.

4.5.5 Torque, parallax adjustment screws. The torque required to rotate the parallax adjustment screw through the range of objective travel shall be measured with standard measuring equipment affixed to the adjustment screw. The measured torque shall not exceed the value specified in 3.6.1 for the temperature indicated.

4.5.6 Sealing. The head assembly and body assembly shall be pressurized to the psig value specified in 3.7. During the time period indicated, the interior pressure shall not decrease in excess of the tolerance specified when checked by suitable measuring equipment. Upon satisfactory completion of the sealing tests, the assemblies shall be flushed with dry nitrogen. The flushing shall continue until the gas bled from the assemblies has a dew-point at least as low as –25° F. at the pressure specified. The assemblies shall then be sealed. This is a post environmental test that shall be performed at +60° to 90° F.

4.5.7 Head assembly. The following tests shall be conducted with the head assembly secured in special testing equipment that shall position the datum planes X and Y, and side Y of the input coupling, as specified in 3.8. The equipment shall be capable of positioning the input coupling to any measured angular position throughout its entire excursion.

4.5.7.1 Collimation, eight power and unity power system. This test shall be performed with special test equipment that shall measure any error in the reflected lines of sight when the incident lines of sight are established as specified in 3.8.1. The reflected lines of sight shall not deviate beyond the tolerance specified in 3.8.1.1 and 3.8.1.2.

4.5.7.2 Plumb travel. This test shall be performed with the aid of an auxiliary telescope containing a vertical reticle line. Position the input coupling to 67 degrees and 30 minutes. The verticle reticle line of the auxiliary telescope shall be adjusted parallel to the image of a calibrated plumb line target. The line of sight (8 power system) shall not deviate more than the tolerance specified in 3.8.2 when the coupling is rotated through the excursion indicated..

4.5.7.3 Elevation accuracy. This test shall be performed with the aid of an auxiliary telescope containing a horizontal reticle line. Direct a line of sight through the 8 power system of the head assembly to a calibrated target appropriate for measuring elevation accuracy of the line of sight. Any measured change in the angular position of the input coupling shall cause the same measured change in the line of sight within the tolerance specified in 8.8.3. Elevation accuracy shall be measured at a minimum of four positions in elevation and three positions in depression starting with the line of sight horizontal. One of these positions shall be at 22 degrees elevation and another at 18 degrees depression.

4.5.7.4 Backlash. This test shall be performed utilizing the target specified in 4.5.7.3. With the backlash excluded for direction of travel, record position of the input coupling and line of sight. Increase or decrease the angular position of the coupling approximately 10 degrees; then, With the backlash included, return the coupling to the recorded position without overtravel and record position of the line of sight. The difference between the two recorded positions of the line of sight is the amount of backlash. Backlash shall not exceed the tolerance specified in 3.8.3.1.

4.5.7.5 Coupling travel. The excursion of the input coupling shall be tested manually, and shall meet the requirements specified in 3.8.4.

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4.5.7.6 Torque, input coupling. The torque required to rotate the input coupling through the specified excursion shall be measured using standard measuring equipment affixed to the coupling. The measured torque shall not exceed the value specified in 3.8.5 for the temperatures indicated in 3.4.2.

4.5.7.7 Resolution (8 power and unity power system). This test shall be performed utilizing an auxiliary telescope of at least 3 power, a resolving power chart (on a target or in a collimator projector) representing the angular subtense for the seconds of are specified for resolution, and a supporting medium to receive this test equipment as well as the item to be tested. The test method and procedure shall be in compliance with the Resolution Test specified in Specification MIL-O-13830. The resolving power chart shall contain 4 sets of lines as shown in Figure 1 of Specification MIL-O-13830. The limit of resolution shall be within the value specified for the unity power and 8 power system in 3.8.6 and 3.9.11, respectively, as determined by viewing through the auxiliary telescope.

4.5.8 Body assembly, 8 power visual. The following tests shall be conducted with the body assembly secured in special testing equipment that shall position datum plane X vertical and reference line Y horizontal. The equipment shall permit a line of sight through the reticle boresight mark of the body assembly to be directed to a collimator type target. The target shall permit accurate measurement of the applicable characteristics of the body.

4.5.8.1 Collimation. Adjust the line of sight through the reticle boresight mark coincident to the intersection of the horizontal and vertical target line image, within the tolerance specified in 3.9.1, by means of the boresight knobs.

4.5.8.2 Elevation knob travel. From the position established in 4.5.8.1, rotate the knob to its clockwise stop then its counterclockwise stop. This knob rotation shall move the reticle boresight point the distance as specified in 3.9.2 when measured on the target.

4.5.8.3 Azimuth knob trarel. Perform this test as specified in 4.5.8.2. Knob travel shall meet the requirements specified in 3.9.3.

4.5.8.4 Reticle shift. This test shall be conducted concurrently with the tests specified in 4.5.8.1. During rotation of the knobs, the reticle boresight point shall not shift perpendicular to the line of sight more than the tolerance specified in 3.9.4.

4.5.8.5 Backlash. With backlash excluded for direction of travel, record position Of knobs and line of sight. Rotate the knob one-half of a revolution: then, with backlash included, return the knob to the recorded position without overtravel and record position of the line of sight. The difference between the two recorded positions of the line of sight is the amount of backlash. Backlash shall not exceed the tolerance specified in 3.9.5.

4.5.8.6 Disengagement of boresight locks. The force required to disengage the boresight lock from the locked position shall be measured with standard measuring equipment. The measured force shall not exceed the value specified for the temperature indicated Slip scales shall be manually tested at the temperatures indicated and shall meet the requirements specified in 3.9.6.

4.5.8.7 Torque, boresight knobs and diopter adjustment. The torque required to rotate the boresight knobs and diopter adjustment shall be measured with standard measuring equipment affixed to the boresight knobs and diopter adjustment. The measured torque of' the boresight knobs and diopter adjustment shall not exceed the value specified in 3.9.7 and 3.9.12.3 at the temperatures indicated in 3.4.2.

4.5.8.8 Reticle illumination. This test shall be conducted with the reticle lamp energized by a 24-volt d-c source. This source shall contain a rheostat for varying reticle illumination for best definition. The reticle mark-

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ings shall meet the requirements specified in 3.9.8.

4.5.8.9 Image tilt. This test shall be performed with the aid of an auxiliary telescope. The vertical reticle line of the auxiliary telescope shall be adjusted parallel to the image of the vertical target line. When the body assembly is placed in front of the auxiliary telescope, the image of the vertical target line shall be parallel to the image of the vertical reticle line of the auxiliary telescope within the tolerance specified in 3.9.9.

4.5.8.10 Parallelism of image and reticle. When one extremity of the longest horizontal reticle line is coincident to the image of a horizontal target line, the reticle line shall be parallel to the target line within the tolerance specified in 3.9.10, as measured on the target.

4.5.8.11 Resolution. The resolution test for body assembly shall be performed in accordance with 4.5.7.7 to meet the requirements of 3.9.11.

4.5.8.12 Eyepiece focus. This test shall be conducted with the aid of a calibrated dioptometer, having a magnification of at least three power, adjusted for best focus of the dioptometer reticle with the dioptometer scale set to zero. Adjust the diopter scale of the body assembly to zero and position the dioptometer at the eyepiece. The reticle of the body assembly shall be in best focus within the tolerance specified in 3.9.12 as measured on the dioptometer scale.

4.5.8.13 Diopter movement. The diopter movement of the eyepiece shall be manually and visually tested for the requirements specified in 3.9.12.1.

4.5.8.14 Backlash. The diopter movement shall be manually and visually tested for backlash. Backlash shall not exceed the tolerance specified in 3.9.12.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8289172. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Periscope, Tank: M31 is primarily intended for use as the gunner's periscope of the primary fire control systems for the 105MM Gun Tanks: M60 and M60A1 and the 90MM Gun Tank: M48A3. The Periscope, Tank: M31C is primarily intended for use in the Combat Engineers Vehicle: T118E1. Both periscopes have an eight power optical system with reticle for laying the main armament on target and a wide range unity power system for observation purposes.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Model of periscope being procured (M31 or M31C).
(c) Selection of applicable levels of preservation, packaging, and packing (see 5.1).
(d) Applicable packaging data sheet number (see 5.1).
(e) Applicable stock number.

6.3 Elevation input coupling. At no time shall the elevation input coupling be allowed to strike hard against the travel stop pin.

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